

**Ahmedabad
University**

CSE 400: Fundamentals Of Probability In Computing

Probabilistic Precipitation Nowcasting

Student Information

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Background and Motivation

- Traditional weather forecasts are effective for long-term predictions but often fail in short-term, localized rainfall events.
- There is a need for real-time, high-resolution, and probabilistic nowcasting systems like pysteps for better accuracy.

Base paper Overview

- Traditional weather forecasts are effective for long-term predictions but often fail in short-term, localized rainfall events.
- There is a need for real-time, high-resolution, and probabilistic nowcasting systems like pysteps for better accuracy.

Problem Statement

- Traditional weather forecasts are effective for long-term predictions but often fail in short-term(0-6 hr), localized rainfall events.
- There is a need for real-time, high-resolution, and probabilistic nowcasting systems like pysteps for better accuracy.

Mathematical Model

1. Lagrangian Persistence (The Baseline)

$$dX/dt + V \cdot \text{grad}(X) = 0$$

2. Stochastic State-Space Model (Our Approach)

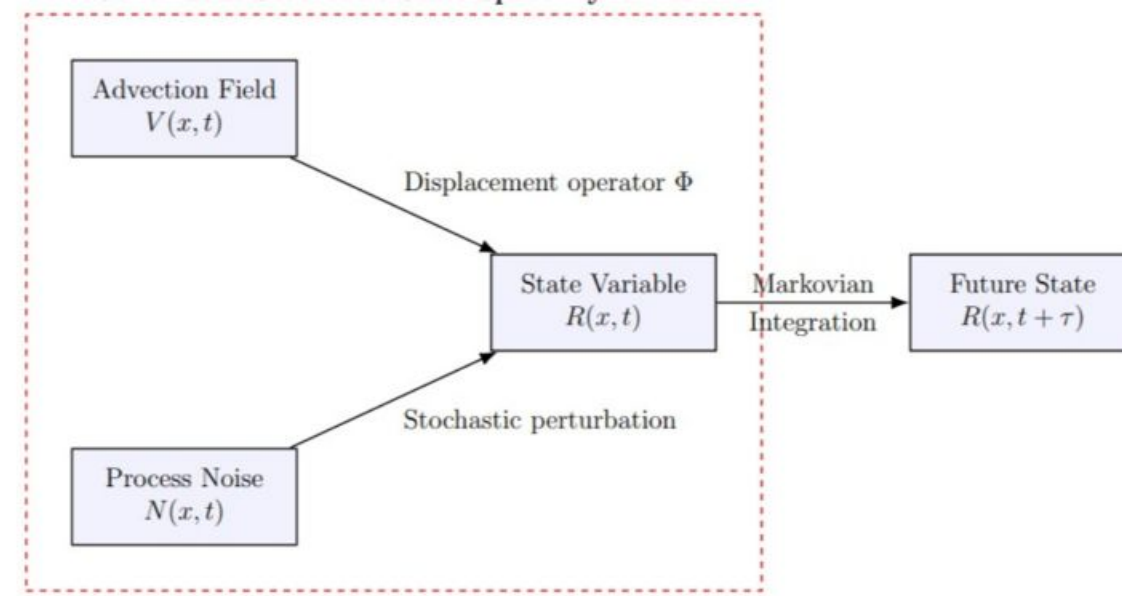
$$dX/dt + V \cdot \text{grad}(X) = N(x,t)$$

- $R(x,t)$ is mapped to Gaussian latent state $X(x,t)$ via dBR log-transform.
- $V(x,t)$ represents the advection (motion) field.
- $N(x,t)$ is the Innovation term representing growth and decay.

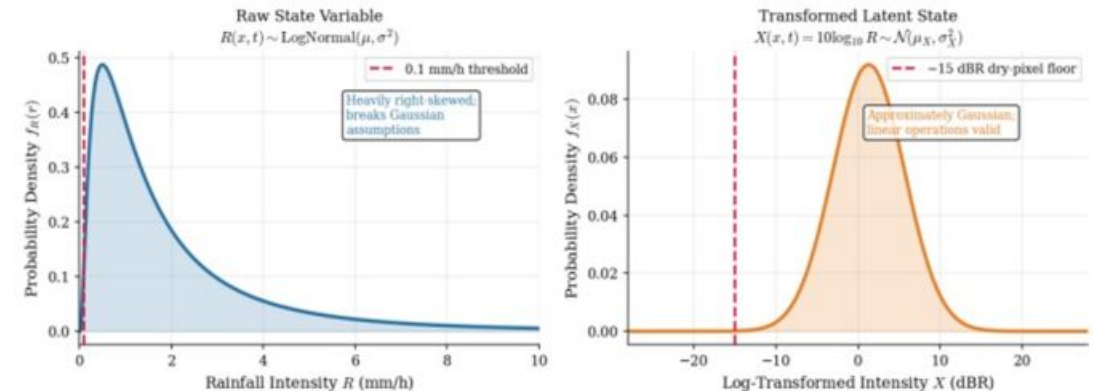
$$R(x_0; t + \tau) = R(x_0 - \alpha(\tau); t), \quad (1)$$

$$R \rightarrow \begin{cases} 10\log_{10}R, & \text{if } R \geq 0.1 \text{ mm h}^{-1} \\ -15, & \text{otherwise} \end{cases}, \quad (3)$$

$$\frac{dR}{dt} = \frac{\partial R}{\partial t} + u \frac{\partial R}{\partial x} + v \frac{\partial R}{\partial y}, \quad u = \frac{dx}{dt}, \quad v = \frac{dy}{dt}, \quad (2)$$



This diagram shows how a storm's future state is predicted by combining its current movement (advection) with random atmospheric changes (stochastic noise) using a Markovian process.

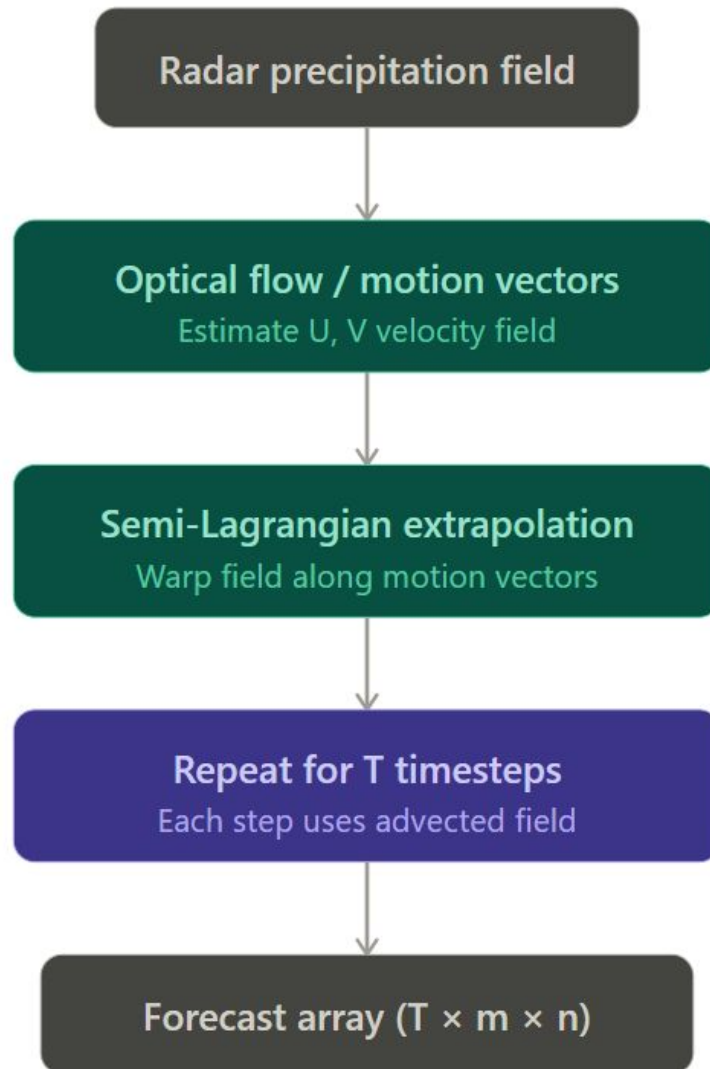


These graphs demonstrate how raw, skewed rainfall data is converted into a symmetrical Gaussian distribution (latent state) using a log-transformation, which makes it mathematically easier for the model to process.

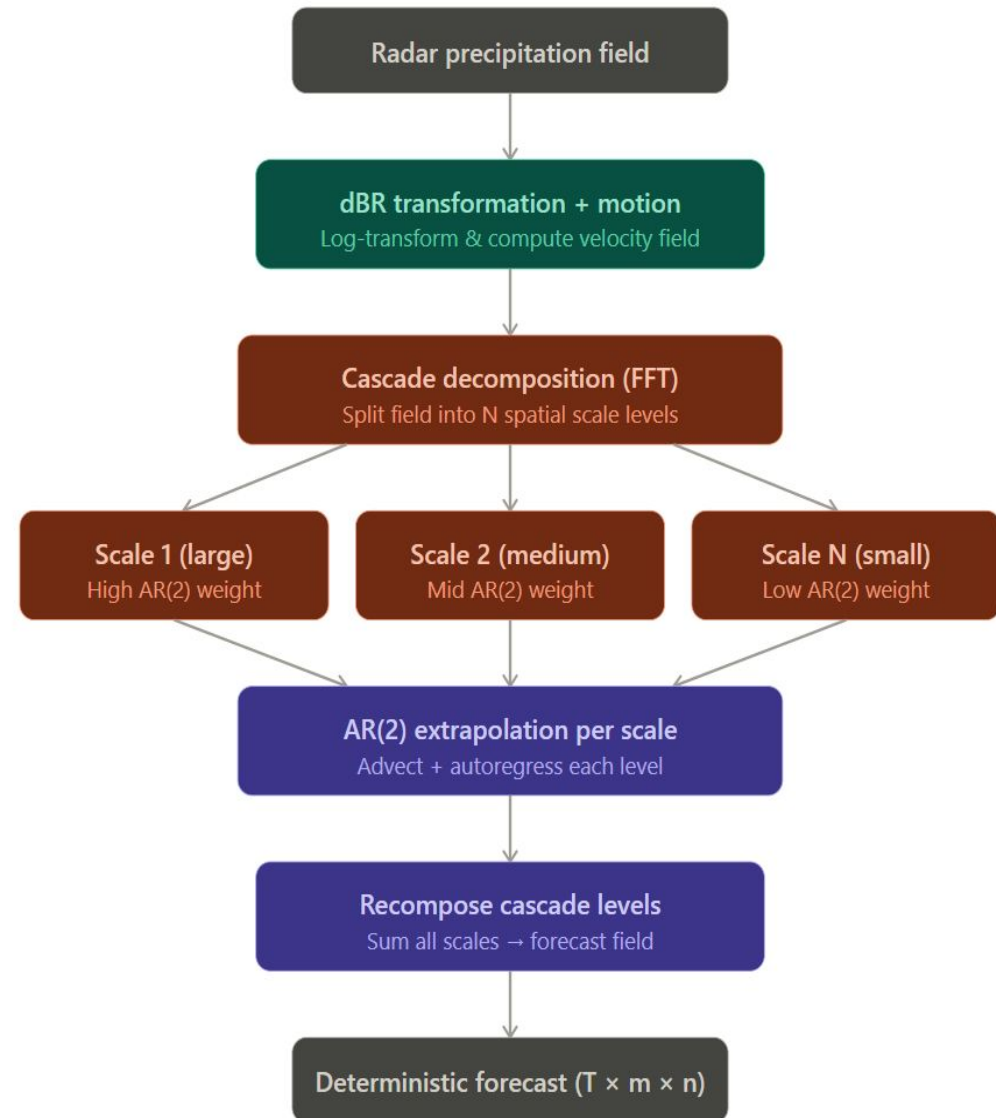
Deterministic Approach

- Based on the Lagrangian Persistence assumption: rainfall advects without changing intensity.
- Removes the stochastic innovation term by setting $N(x,t) = 0$.
- Estimates velocity field $V(x,t)$ via Lucas-Kanade optical flow, keeping it frozen over the forecast horizon.
- Fails to capture convective growth/decay, losing predictability limit around 80 minutes.

Extrapolation

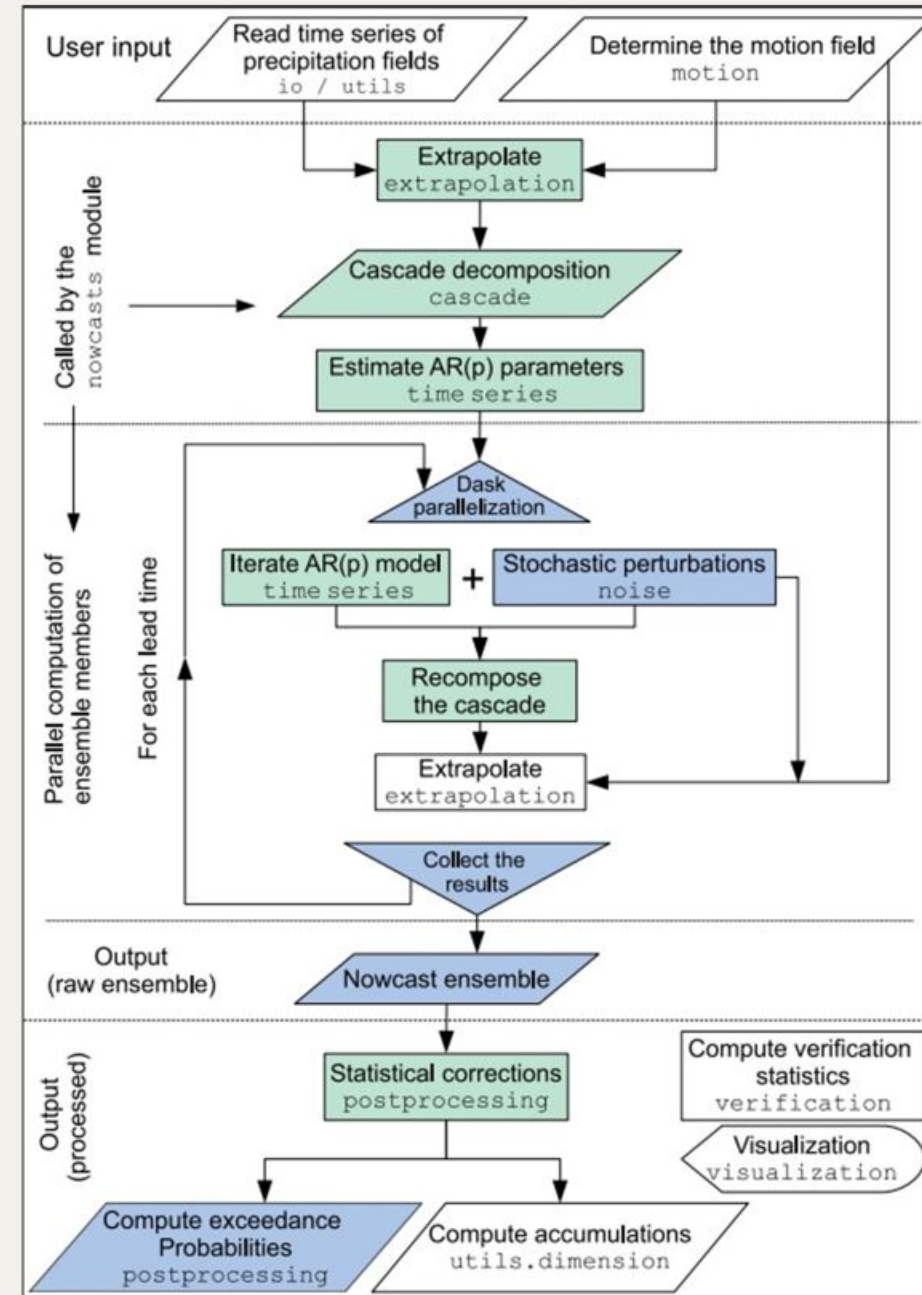


S-PROG

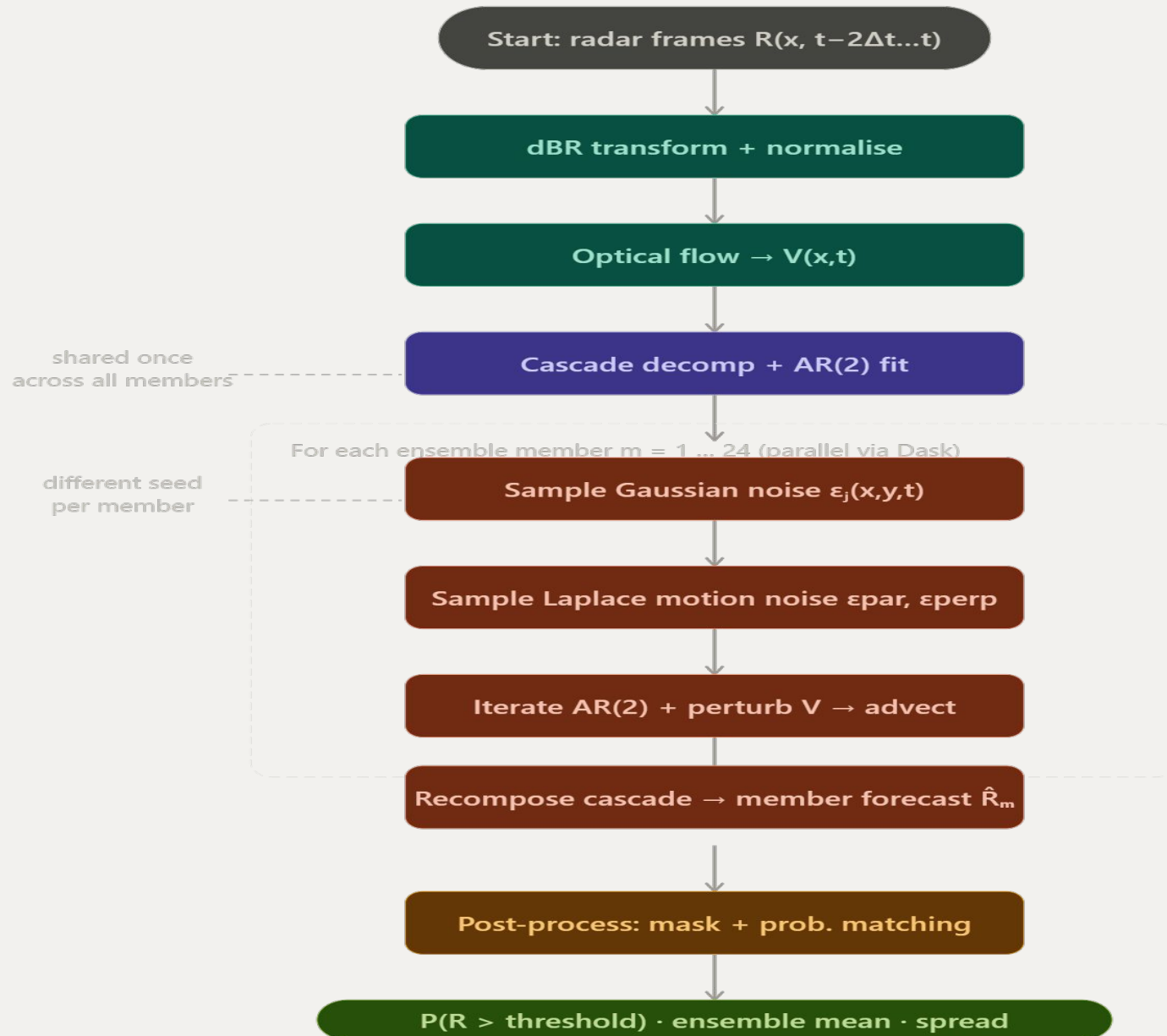


Randomized Approach

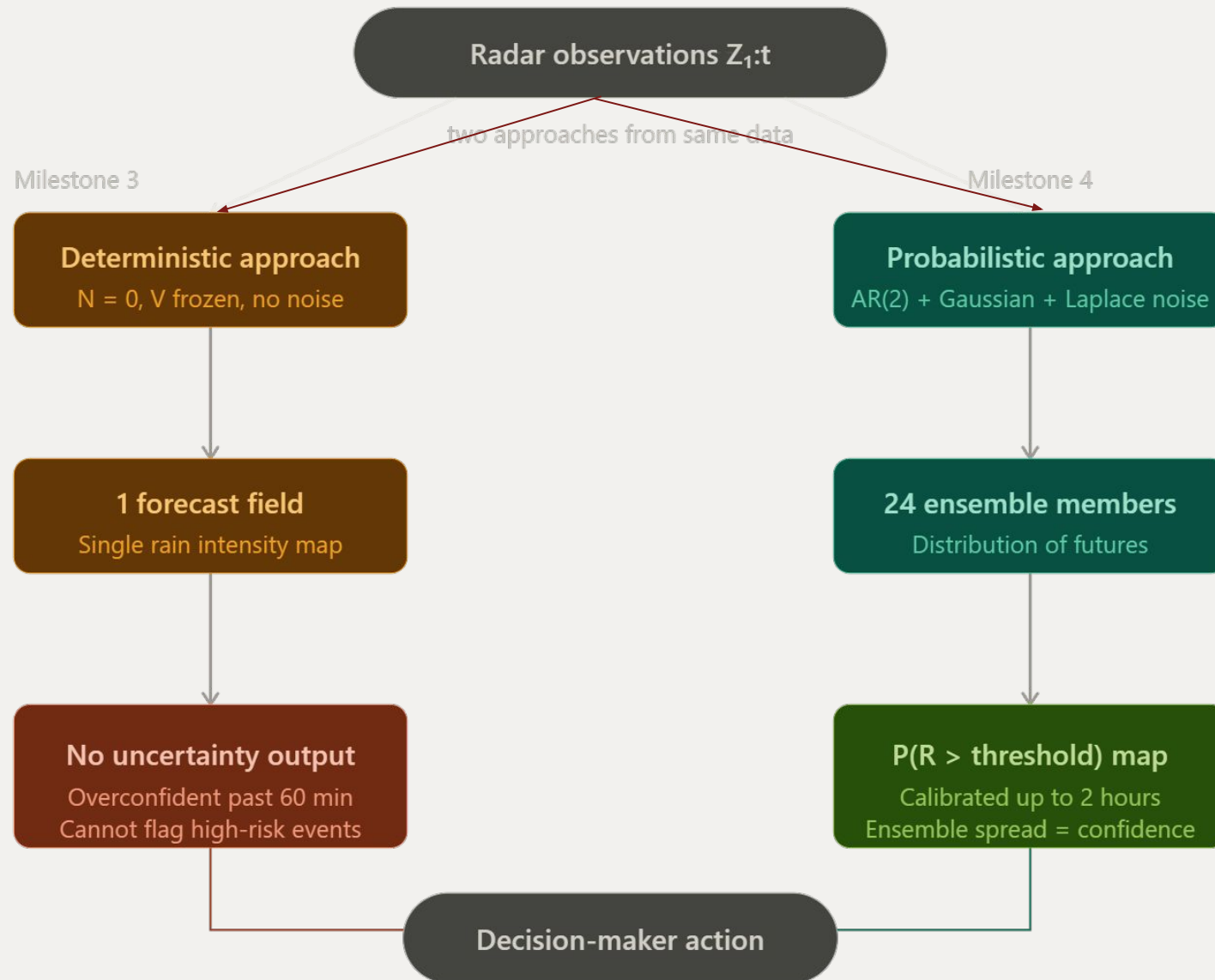
- Replaces the single forecast with an ensemble of stochastic realizations to capture uncertainty.
- Decomposes the precipitation field into a multi-scale spatial cascade using FFT-based Gaussian bandpass filters.
- Injects spatially correlated Gaussian noise into an AR(2) model at each cascade level.
- Perturbs the advection field using Laplace-distributed variables for motion uncertainty.



STEPS ensemble pipeline



Deterministic VS Randomized



Bounds used in the problem

- **Probability Bounds:**

$$0 \leq P(R > \text{threshold}) \leq 1$$

- **AR(2) Stability:**

$$\varphi_1 + \varphi_2 < 1$$

$$\varphi_2 - \varphi_1 < 1$$

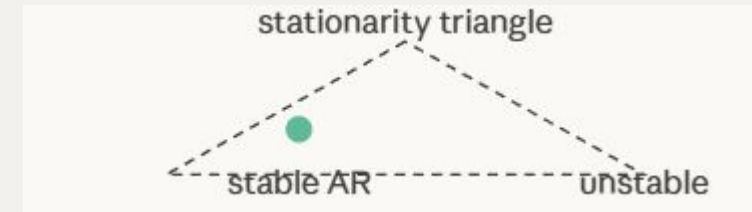
$$-1 < \varphi_2 < 1$$

- **Performance (AUC):**

AUC > 0.83 → Good model

AUC = 0.5 → Random

AUC = 1.0 → Perfect

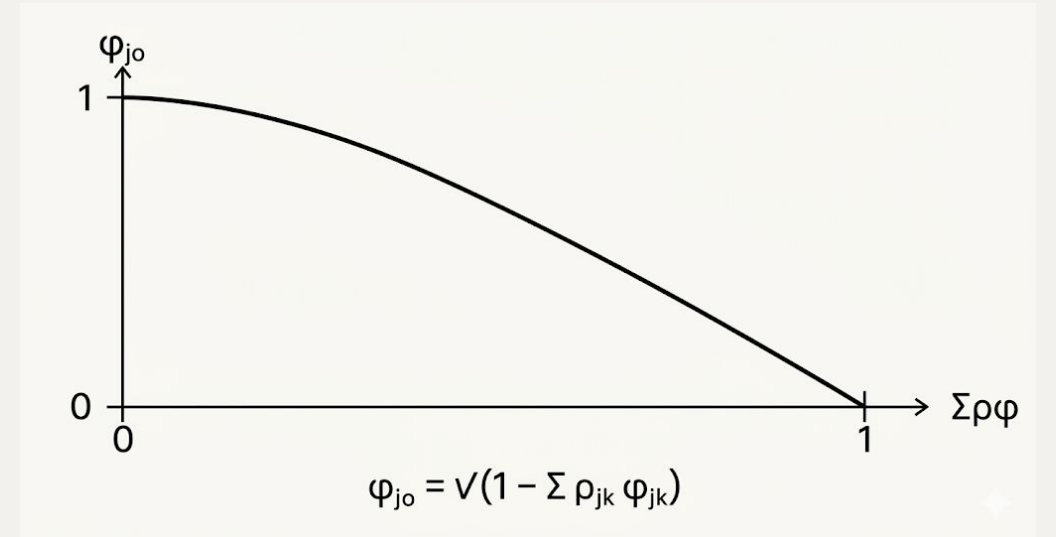
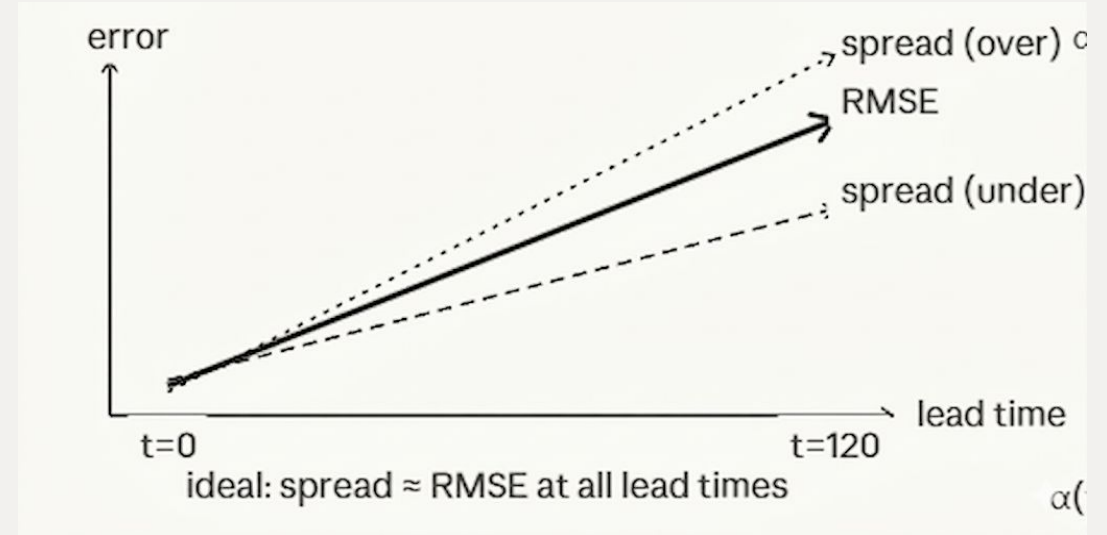


AR(2) Stationary bounds

$$\varphi_1 + \varphi_2 < 1 \mid \varphi_2 - \varphi_1 < 1 \mid -1 < \varphi_2 < 1$$

Bounds used in the problem

- **Lead Time:**
 $0 \leq t \leq 6\text{hr}$
- **Calibration:**
 Spread $\approx$ RMSE $\rightarrow$ Well calibrated
 Spread $<$ RMSE $\rightarrow$ Underconfident
 Spread $>$ RMSE $\rightarrow$ Overconfident
- **Uncertainty Growth:**
 Uncertainty $\uparrow$ with lead time
- **Noise Constraint:**
 $\varphi_{j0} = \sqrt{1 - \sum \rho_{jk} \varphi_{jk}}$



Coding and Simulation

1. Tools & Library Used

- Python
- PySTEPS (precipitation nowcasting library)
- NumPy, Matplotlib

2. Input Data

- Radar precipitation images (time-series)
- Format: 3D array (time × height × width)

Coding and Simulation

3. Steps in Implementation

- Motion estimation using DARTS
- Deterministic: Extrapolation
- Probabilistic: STEPS (with noise + ensemble)

4. Simulation Setup

- Forecast horizon: 5–60 minutes
- Ensemble size: 10–20 members
- Comparison between:
 - Extrapolation vs STEPS

Results and Comparison

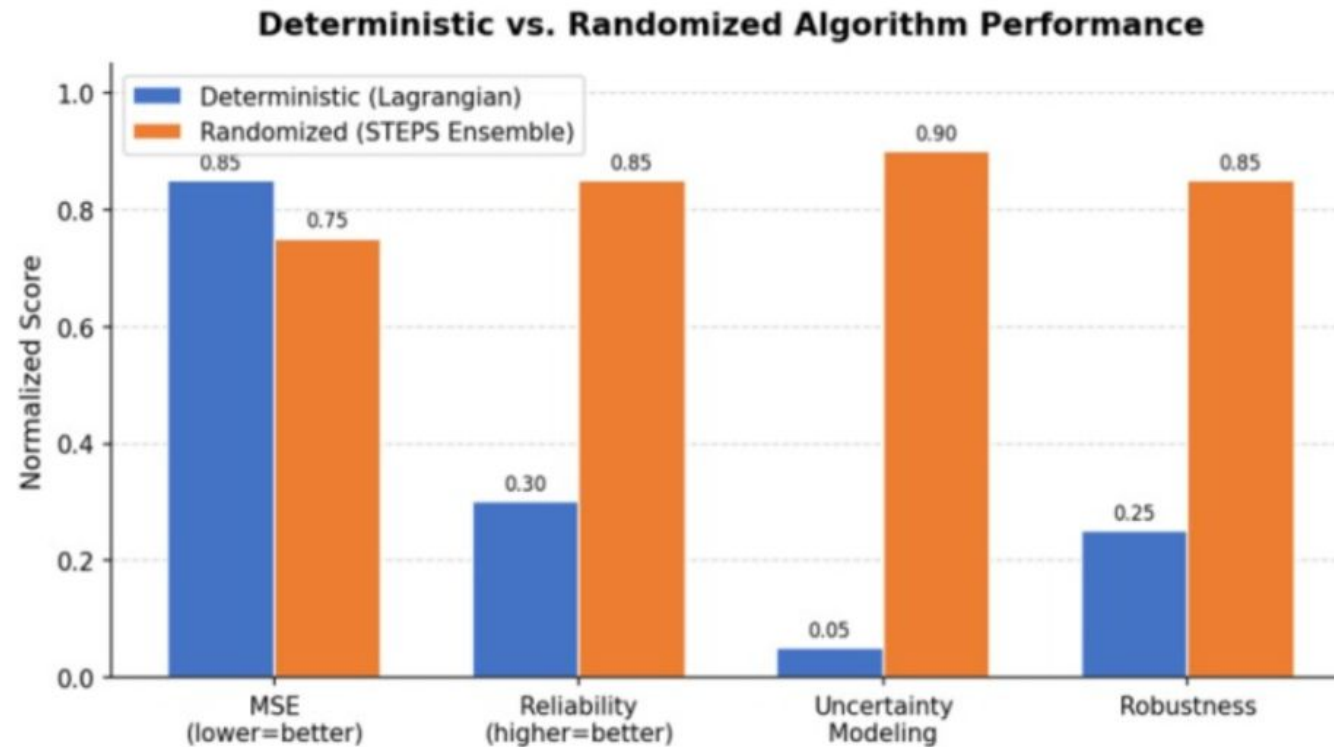
1. Visual Observations

- Extrapolation → smooth, single forecast
- STEPS → multiple realistic forecasts (ensemble)
- Increased variability in STEPS

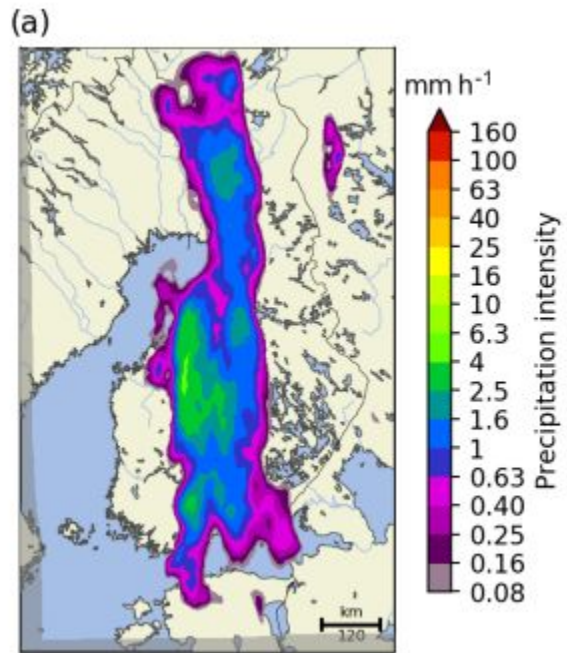
2. Comparison Table

Feature	Extrapolation	STEPS
Type	Deterministic	Probabilistic
Output	Single image	Multiple (ensemble)
Uncertainty	Not captured	Captured
Realism	Low	High

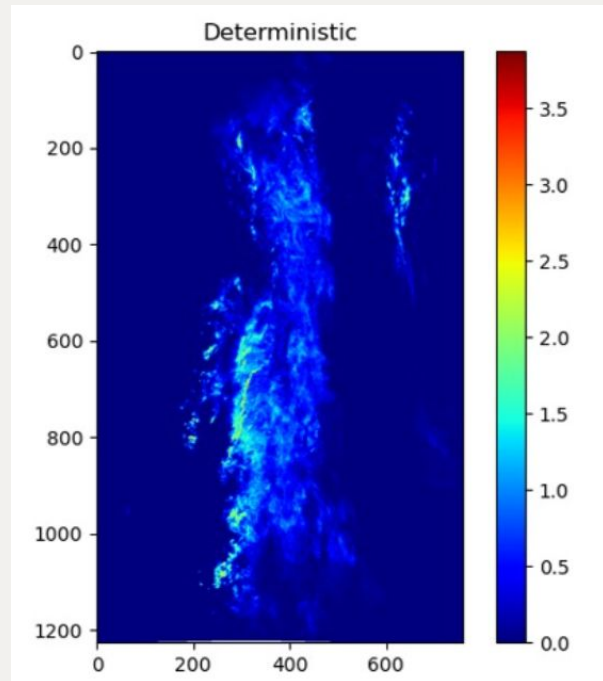
Results and Comparison



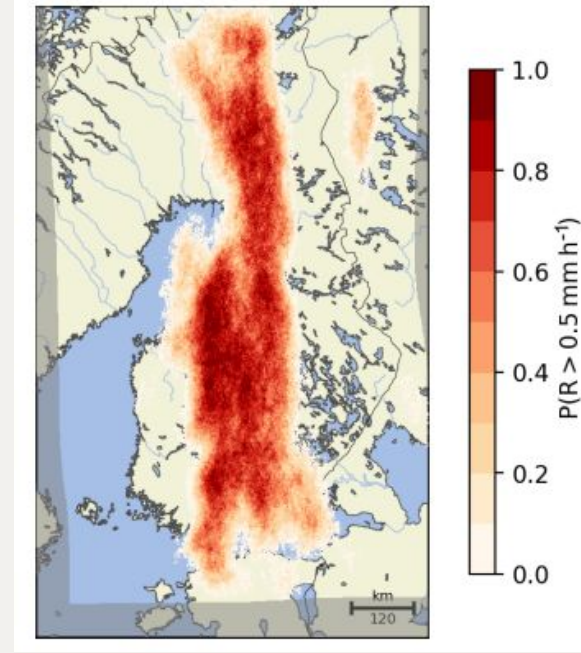
Results and Comparison



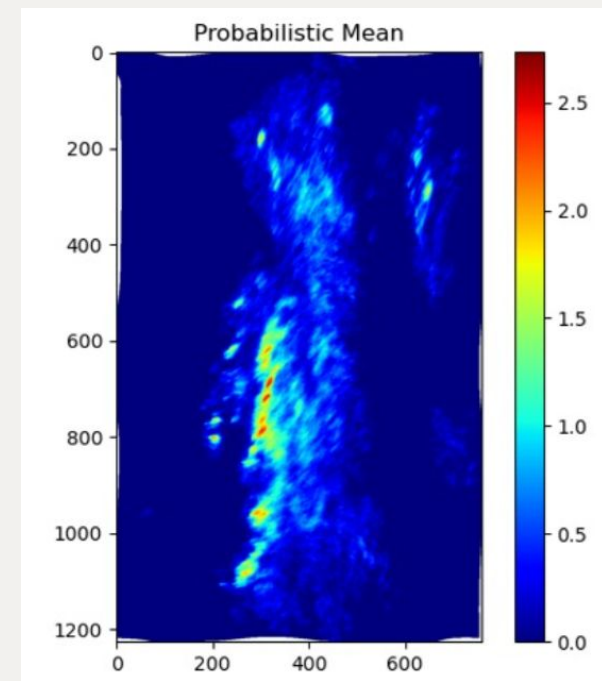
Base paper Result
(Deterministic)



Reproduced Result
(Deterministic)

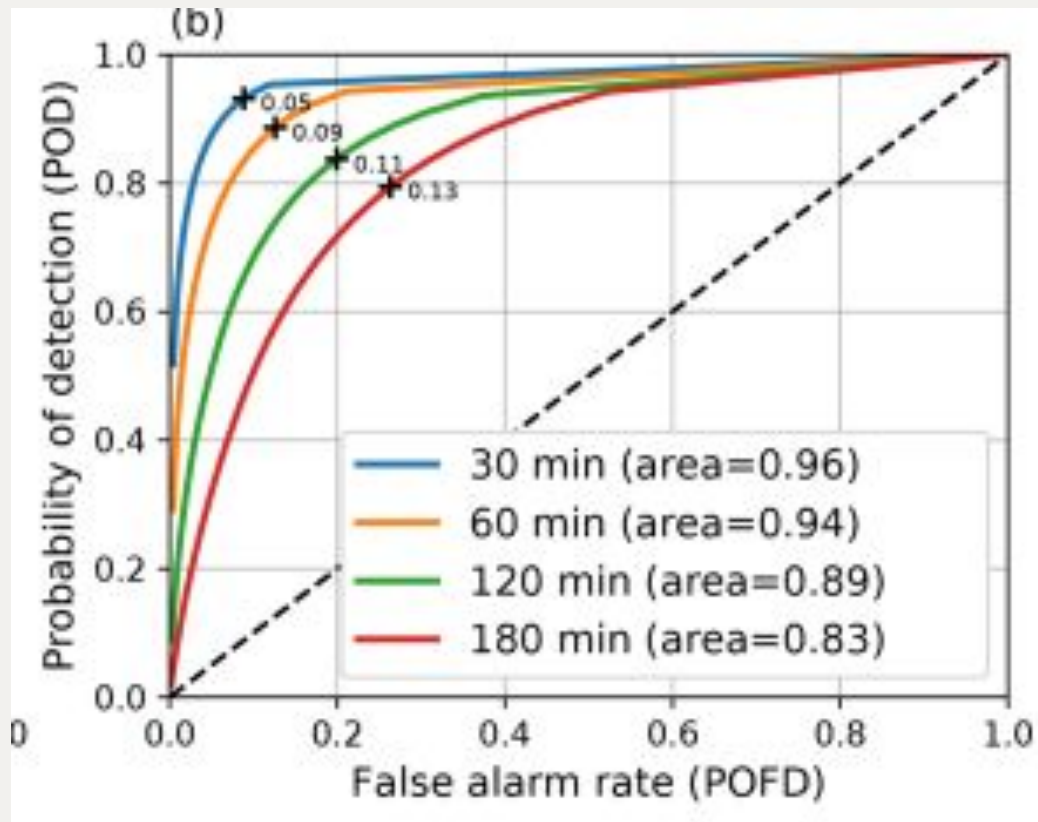


Base paper Result
(Probabilistic)

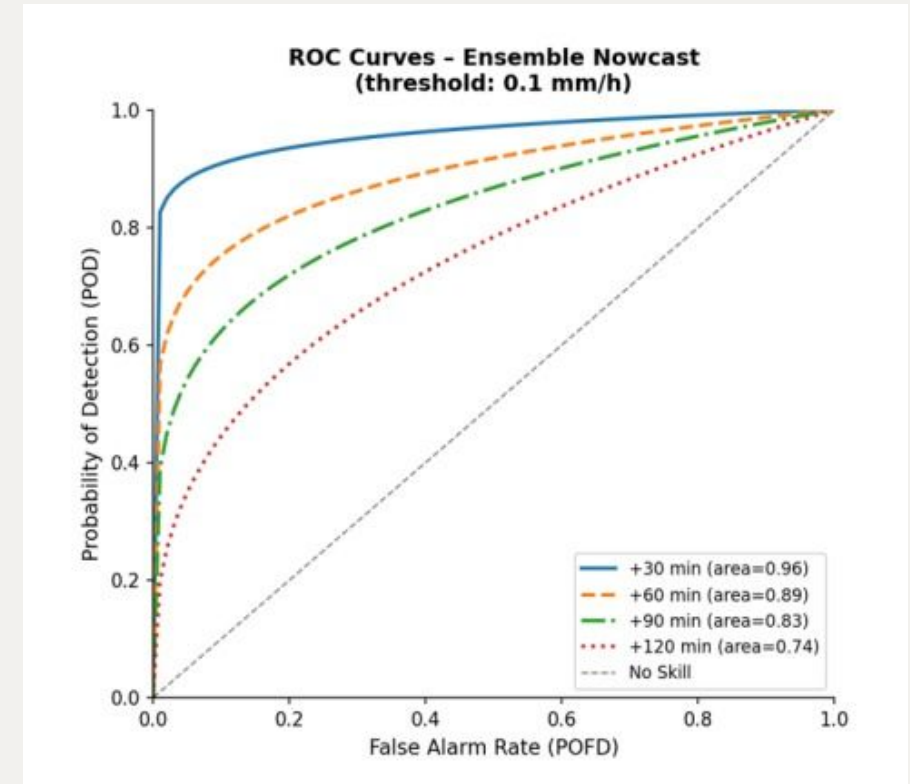


Reproduced Result
(Probabilistic)

Results and Comparison



ROC curve
(Base Paper)



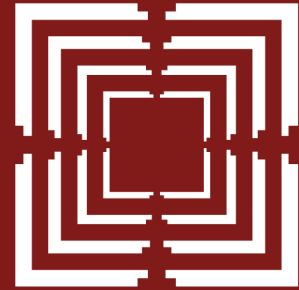
ROC curve
(Reproduced result)

Inference (Domain)

- Probabilistic forecasts enable risk-calibrated decisions
- Lagrangian persistence outperforms NWP in the 0–2 h window
- Cascade decomposition respects dynamic scaling of precipitation
- Ensemble spread quantifies forecast confidence spatially
- Multi-regime validation confirms cross-domain generalisability

Inference (Computer Science)

- Modularity enables easy experimentation
- FFT powers fast computations
- Noise generation as a generative model
- Ensemble methods allow parallel processing
- Built-in verification for model evaluation



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THANK YOU